TASK 4: INTEGRATED DIAGNOSTIC CAPSTONE AND PROFESSIONAL DISCUSSION

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 10 | v1.3

TIME	EVIDENCE	PROGRAMME POINT
90 minutes (adapt to workplace conditions)	**Practical | Knowledge | Documentation**	**Weeks 129-135 | Months 34-35**

Your task

Complete a developmental capstone diagnosis on a heavy-vehicle system, justify the evidence and repair decision, verify the outcome, and explain professional behaviours and learning in a structured discussion.

What you must do

1. Confirm the symptom and create a risk-controlled diagnostic plan.
2. Use the focused checklist and suitable equipment to collect valid evidence.
3. Identify root cause, plan/complete authorised action and verify the result.
4. Produce a clear job card, customer/operator explanation and evidence references.
5. Reflect on ownership, teamwork, limitations and next development action.

What good evidence looks like

SAFE CONTROL

The activity is developmental training, not an EPA decision; work beyond authority must be stopped and referred through the approved process.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- Heavy vehicle or integrated diagnostic rig
- Manufacturer data and company procedures
- Technical and diagnostic equipment appropriate to fault
- Job card, evidence pack and discussion prompts

SOW ALIGNMENT

Block 10 - Final Consolidation, Corrective Learning and EPA Readiness Training

KSB evidence focus: S11, B1; integrated evidence across K1-K11, S1-S11, B1-B3

Task reference: HV_L3_Block_10_Task_4_Integrated_Diagnostic_Capstone_and_Professional_Discussion

FOCUSED DIAGNOSTIC INSPECTION

Checklist | Complete the relevant checks for the selected fault

Apprentice | **Vehicle / trailer ID**

Date | **Job card / evidence ref**

Selected complaint / system

Use the checks that apply to the chosen fault. Tick Done or N/A and record measured evidence below.

Diagnostic inspection and decision checks

Item and component	Look for	Done	N/A
D1 Confirm driver symptom and operating conditions	Complaint, history and when it occurs	☐	☐
D2 Secure vehicle and control stored energy	Chocks, support, isolation and authority	☐	☐
D3 Inspect the affected system visually	Condition, wear, leaks, connections	☐	☐
D4 Check current technical / DVSA criteria	Manual, limits, diagram and test method	☐	☐
D5 Check base supply or mechanical condition	Air, fluid, tyres, linkage or supply	☐	☐
D6 Collect relevant scan or measured data	DTC, live data, pressure or voltage	☐	☐
D7 Compare actual results with the criterion	Record units and test conditions	☐	☐
D8 Identify cause and competing explanations	Evidence supports the diagnosis	☐	☐
D9 Agree repair, referral or stop decision	Authorisation and safety implications	☐	☐
D10 Verify result by an approved method	Repeat test, function, warning status	☐	☐
D11 Complete job card and explanation	Findings, limits and next action	☐	☐

Selected-system prompts | complete only where relevant

☐ Brake imbalance: wear, actuator, air/EBS and approved performance evidence

☐ Heavy steering: tyres, linkage, assistance and approved functional evidence

Manual / criterion ref

Actual values, units and conditions

Cause, action and verification ref

Use the core task pages for full findings, developmental PASS/REFER, feedback and signatures.

TASK 4: INTEGRATED DIAGNOSTIC CAPSTONE AND PROFESSIONAL DISCUSSION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Symptom / operating condition		Primary test / expected value	
Actual result / evidence		Root cause	
Repair / referral action		Verification result	
Customer explanation ref		Development action	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why was your first test the best next test?

2. How did you confirm cause rather than correlation?

Knowledge check - continued

3. Explain how safety, time and cost affected your decision.

4. How did you respond when something did not go to plan?

5. Which evidence best demonstrates readiness and what remains to develop?

Assessor/Workplace Mentor observation

- ☐ Creates and follows a safe diagnostic strategy.
- ☐ Collects valid evidence and identifies root cause.
- ☐ Verifies work and produces an accurate record.
- ☐ Communicates technical decisions clearly.
- ☐ Demonstrates ownership, integrity and reflective development.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ☐ PASS
- ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

FAULT SCENARIOS AND EVIDENCE

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 10 | Task 4

How to use this sheet

Work through the listed scenarios during the block using a vehicle, safe fault rig or supported case study. Record which scenario was used for this task and reference the actual measurements and assessor observation on the existing practical record. A discussion or case study alone is not evidence of practical competence.

1. Brake imbalance

Checks: Use approved brake performance evidence alongside friction, actuator, adjustment, air and EBS checks; identify the affected axle/side.

Evidence: Record brake results, cause, repair/referral and approved performance verification.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

2. Steering heavy

Checks: Check tyres, steering linkage/geometry, hydraulic or electric assistance, fluid/pressure and system-specific controls.

Evidence: Record conditions, measurements, diagnosis and decision on safe movement.

Completed as: [ ] Vehicle/rig practical [ ] Supported case study [ ] Further practice needed

Date / evidence or job-card reference: _____

Scenario selected for the original task assessment: _____

For any safety-critical fault, stop or control the work and follow the manufacturer, DVSA and workplace procedures. Use the original task pages for findings, feedback, PASS/REFER and signatures.

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.